

License

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For full details, please see the [LICENSE](#) file in the repository.

Go

```
go get github.com/mikhaildadaev/uuid
```

`uuid` `v1.26.11.`

Run Test

```
go test ./...
go test -bench=. ./...
go test -cover ./...
go test -race ./...
```

Key Features

- UUIDv1..v8 Null — UUID SQL null UUID
- — null UUID
- — POSIX
- — Binary JSON Text /
- SQL — database/sql Scan/Value
- —

Limits

- V2: `posix` `0..255` ;
- V3/V5: `name` `1..512`
- V8: `node` `0..16383` ;

0

512

name

null UUID

API / Core /

UUID

—

SQL

NULL

V7

V8

NewV4

NewV7

NewNull

SQL

NULL

UUID

```
un := uuid.NewNull()  
fmt.Println(un.IsZero())  
fmt.Println(un.String())
```

```
true  
00000000-0000-0000-0000-000000000000
```

NewV1

MAC

UUID

1

```
u1 := uuid.NewV1()  
fmt.Println(u1.Version())
```

```
1
```

NewV2

UUID

2

DCE

MAC

POSIX

UID/GID

```
u2 := uuid.NewV2(posix)
fmt.Println(u2.Version())
```

2

NewV3

MD5

UUID

3

```
u3 := uuid.NewV3(uuid.NameSpaceDNS, name)
fmt.Println(u3.Version())
```

3

NewV4

UUID

4

```
u4 := uuid.NewV4()
fmt.Println(u4.Version())
```

4

NewV5

SHA-1

UUID

5

```
u5 := uuid.NewV5(uuid.NamespaceDNS, name)
fmt.Println(u5.Version())
```

5

NewV6

	MAC	MAC	UUID	6
UUIDv1				

```
u6 := uuid.NewV6()
fmt.Println(u6.Version())
```

6

NewV7

UUID	7	Unix
------	---	------

```
u7 := uuid.NewV7()
fmt.Println(u7.Version())
```

7

NewV8

UUID	8
------	---

```
u8 := uuid.NewV8(node)
fmt.Println(u8.Version())
```

8

API / Core /

	UUID	NullUUID	String	Version	
Timestamp	Node	Sequence		UUID	V1
V8					

NULLUUID

IsZero

NullUUID	SQL NULL	UUID	true
----------	----------	------	------

```
var nu uuid.NullUUID
fmt.Println(nu.IsZero())
nu.Scan("01968727-8c7e-8000-87cb-bdba4f634d9f")
fmt.Println(nu.IsZero())
```

true
false

String

UUID	8-4-4-4-12	00000000-0000-0000-0000-
000000000000		

```
var nu uuid.NullUUID
fmt.Println(nu.String())
nu.Scan("01968727-8c7e-8000-87cb-bdba4f634d9f")
fmt.Println(nu.String())
```

```
00000000-0000-0000-0000-000000000000
01968727-8c7e-8000-87cb-bdba4f634d9f
```

Validate

NullUUID
UUID

UUID

nil

SQL

```
var nu uuid.NullUUID
fmt.Println(nu.Validate())
nu.Scan("01968727-8c7e-8000-87cb-bdba4f634d9f")
fmt.Println(nu.Validate())
```

```
<nil>
<nil>
```

UUID

Bytes

UUID 16

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
fmt.Printf("%x\n", uu.Bytes())
```

019687278c7e800087cbbdba4f634d9f

Equal

UUID

UUID

true

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
other := uu
fmt.Println(uu.Equal(other))
```

true

Info

UUID

POSIX

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
fmt.Println(uu.Info())
```

```
UUID: 01968727-8c7e-8000-87cb-bdba4f634d9f
VAR.: RFC4122
VER.: 8
FORM: TTTTTTTT-TTTT-8SSS-VNNN-RRRRRRRRRRRR
INFO: TIME (1-milliseconds interval since 1970-01-01) + SEQUENCE (0-4095) + NOI
TIME: 1746024238206 [2025-04-30 14:43:58.206 +0000 UTC]
SEQ.: 0
NODE: 1995
RAND: bdba4f634d9f
```

IsZero

UUID 16 true

```
uu := uuid.NewNull()
fmt.Println(uu.IsZero())
```

```
true
```

Node

UUID V1 V2 V6 V8 0

```
uuidNode := 1995
u8 := uuid.NewV8(uuidNode)
fmt.Println(u8.Node())
```

```
1995
```

Posix

UUID V2 POSIX UID/GID 0

```
uuidPosType := 0
uuidPosValue := 501
u2 := uuid.NewV2(uuidPosType, uuidPosValue)
fmt.Println(u2.Posix())
```

```
UID 501
```

Sequence

UUID V1 V2 V6 V7 0

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
fmt.Println(uu.Sequence())
```

0

String

UUID

8-4-4-4-12

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
fmt.Println(uu.String())
```

01968727-8c7e-8000-87cb-bdba4f634d9f

Timestamp

UUID

V1

V2

V6

V7

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
fmt.Println(uu.Timestamp())
```

1746024238206

Validate

UUID

UUID

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"  
uu, err := uuid.Parse(uuidV8String)  
if err != nil {  
    fmt.Println(err)  
}  
fmt.Println(uu.Validate())
```

<nil>

Variant

UUID

UUID

1 RFC 4122

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"  
uu, err := uuid.Parse(uuidV8String)  
if err != nil {  
    fmt.Println(err)  
}  
fmt.Println(uu.Variant())
```

1

Version

UUID

1 8

null UUID

0

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"  
uu, err := uuid.Parse(uuidV8String)  
if err != nil {  
    fmt.Println(err)  
}  
fmt.Println(uu.Version())
```

8

API / Marshal /

	UUID	Binary	Json	Text
UUID	— V1	V8	Null	

NULLUUID

MarshalBinary

NullUUID	16
----------	----

```
var nu uuid.NullUUID
data, err := nu.MarshalBinary()
if err != nil {
    fmt.Println(err)
}
fmt.Printf("%x\n", data)
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
nu.Scan(uuidV8String)
data, err = nu.MarshalBinary()
if err != nil {
    fmt.Println(err)
}
fmt.Printf("%x\n", data)
```

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MarshalJson

NullUUID	Json	null
----------	------	------

```

var nu uuid.NullUUID
data, err := nu.MarshalJson()
if err != nil {
    fmt.Println(err)
}
fmt.Println(string(data))
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
nu.Scan(uuidV8String)
data, err := nu.MarshalJson()
if err != nil {
    fmt.Println(err)
}
fmt.Println(string(data))

```

```

null
"01968727-8c7e-8000-87cb-bdba4f634d9f"

```

MarshalText

NullUUID

```

var nu uuid.NullUUID
data, err := nu.MarshalText()
if err != nil {
    fmt.Println(err)
}
fmt.Println(string(data))
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
nu.Scan(uuidV8String)
data, err := nu.MarshalText()
if err != nil {
    fmt.Println(err)
}
fmt.Println(string(data))

```

```

01968727-8c7e-8000-87cb-bdba4f634d9f

```

UnmarshalBinary

```

var nu uuid.NullUUID
data := []byte{}
if err := nu.UnmarshalBinary(data); err != nil {
    fmt.Println(err)
}
fmt.Println("Valid:", nu.Valid)
fmt.Println("UUID:", nu.UUID)
uuidV8Binary := []byte{0x01, 0x96, 0x87, 0x27, 0x8c, 0x7e, 0x80, 0x00, 0x87, 0x00}
err := nu.UnmarshalBinary(uuidV8Binary)
if err != nil {
    fmt.Println(err)
}
fmt.Println("Valid:", nu.Valid)
fmt.Println("UUID:", nu.UUID)

```

```

Valid: false
UUID: 00000000-0000-0000-0000-000000000000
Valid: true
UUID: 01968727-8c7e-8000-87cb-bdba4f634d9f

```

UnmarshalJson

Json	NullUUID	null	SQL NULL	UUID
------	----------	------	----------	------

```

var nu uuid.NullUUID
data := []byte(`null`)
if err := nu.UnmarshalJson(data); err != nil {
    fmt.Println(err)
}
fmt.Println("Valid:", nu.Valid)
fmt.Println("UUID:", nu.UUID)
uuidV8Json := []byte(`"01968727-8c7e-8000-87cb-bdba4f634d9f"`)
err := nu.UnmarshalJson(uuidV8Json)
if err != nil {
    fmt.Println(err)
}
fmt.Println("Valid:", nu.Valid)
fmt.Println("UUID:", nu.UUID)

```

```
Valid: false
UUID: 00000000-0000-0000-0000-000000000000
Valid: true
UUID: 01968727-8c7e-8000-87cb-bdba4f634d9f
```

UnmarshalText

NullUUID

SQL NULL

```
var nu uuid.NullUUID
data := []byte("")
if err := nu.UnmarshalText(data); err != nil {
    fmt.Println(err)
}
fmt.Println("Valid:", nu.Valid)
fmt.Println("UUID:", nu.UUID)
uuidV8Text := []byte("01968727-8c7e-8000-87cb-bdba4f634d9f")
err := nu.UnmarshalText(uuidV8Text)
if err != nil {
    fmt.Println(err)
}
fmt.Println("Valid:", nu.Valid)
fmt.Println("UUID:", nu.UUID)
```

```
Valid: false
UUID: 00000000-0000-0000-0000-000000000000
Valid: true
UUID: 01968727-8c7e-8000-87cb-bdba4f634d9f
```

UUID

MarshalBinary

UUID 16

```

uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
data, err := uu.MarshalBinary()
if err != nil {
    fmt.Println(err)
}
fmt.Printf("%x\n", data)

```

```
019687278c7e800087cbbdba4f634d9f
```

MarshalJson

UUID 8-4-4-4-12 Json

```

uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
data, err := uu.MarshalJson()
if err != nil {
    fmt.Println(err)
}
fmt.Println(string(data))

```

```
"01968727-8c7e-8000-87cb-bdba4f634d9f"
```

MarshalText

UUID 32

```

uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
data, err := uu.MarshalText()
if err != nil {
    fmt.Println(err)
}
fmt.Println(string(data))

```

```
01968727-8c7e-8000-87cb-bdba4f634d9f
```

UnmarshalBinary

16

UUID

16

```

var uu uuid.UUID
uuidV8Binary := []byte{0x01, 0x96, 0x87, 0x27, 0x8c, 0x7e, 0x80, 0x00, 0x87, 0x
err := uu.UnmarshalBinary(uuidV8Binary)
if err != nil {
    fmt.Println(err)
}
fmt.Println(uu.String())

```

```
01968727-8c7e-8000-87cb-bdba4f634d9f
```

UnmarshalJson

Json

UUID

```

var uu uuid.UUID
uuidV8Json := []byte(`"01968727-8c7e-8000-87cb-bdba4f634d9f"`)
err := uu.UnmarshalJson(uuidV8Json)
if err != nil {
    fmt.Println(err)
}
fmt.Println(uu.String())

```

01968727-8c7e-8000-87cb-bdba4f634d9f

UnmarshalText

32

UUID

```
var uu uuid.UUID
uuidV8Text := []byte("01968727-8c7e-8000-87cb-bdba4f634d9f")
err := uu.UnmarshalText(uuidV8Text)
if err != nil {
    fmt.Println(err)
}
fmt.Println(uu.String())
```

01968727-8c7e-8000-87cb-bdba4f634d9f

API / SQL /

UUID

database/sql

Scan

Value

UUID

SQL

NULL

NULLUUID

Scan

UUID

sql.Scanner

nil

SQL NULL

UUID

/

UUID

Valid

true

NULL

false

```

var nu uuid.NullUUID
if err := nu.Scan(nil); err != nil {
    fmt.Println(err)
}
fmt.Println("Valid:", nu.Valid)
fmt.Println("UUID:", nu.UUID)
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
if err := nu.Scan(uuidV8String); err != nil {
    fmt.Println(err)
}
fmt.Println("Valid:", nu.Valid)
fmt.Println("UUID:", nu.UUID)

```

```

Valid: false
UUID: 00000000-0000-0000-0000-000000000000
Valid: true
UUID: 01968727-8c7e-8000-87cb-bdba4f634d9f

```

Value

UUID	driver.Valuer	Valid	false	nil	Valid
true	UUID				

```

var nu uuid.NullUUID
value, _ := nu.Value()
fmt.Println(value)
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
nu.Scan(uuidV8String)
value, _ = nu.Value()
fmt.Println(value)

```

```

<nil>
01968727-8c7e-8000-87cb-bdba4f634d9f

```

UUID

Scan

`sql.Scanner`

UUID—

nil *NULL*

```
var uu uuid.UUID
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
err := uu.Scan(uuidV8String)
if err != nil {
    fmt.Println(err)
}
fmt.Println(uu.String())
```

01968727-8c7e-8000-87cb-bdba4f634d9f

Value

`driver.Valuer`

UUID

UUID

nil

```
uuidV8String := "01968727-8c7e-8000-87cb-bdba4f634d9f"
uu, err := uuid.Parse(uuidV8String)
if err != nil {
    fmt.Println(err)
}
value, err := uu.Value()
if err != nil {
    fmt.Println(err)
}
fmt.Println(value)
```

01968727-8c7e-8000-87cb-bdba4f634d9f